

$\mathcal{W}$

$\mathbf{t}$

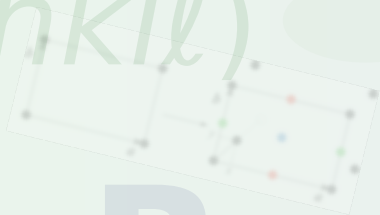
$(hkil)$



230 space groups

$(hkil)$

$\det \mathbf{R} = \pm 1$



$\mathbf{R}$

$(\mathbf{R}|\mathbf{t})$

$$\mathbf{r}' = \mathbf{R}\mathbf{r} + \mathbf{t}$$



32 point groups

$$\tilde{\mathbf{r}}' = \mathcal{W}\tilde{\mathbf{r}}$$